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0.1 Exersice

Exercise 0.1.1 ▸ Calculate

Let V be a vector space of finite dimension with basis $(e_1, \dots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \dots, e_n^*)$. Please prove in two different ways that for every $(\alpha, \beta) \in \Lambda^p V^* \times \Lambda^q V^*$,

$$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha.$$

One way is to use definition, the other is to calculate in basis using the next exercise.

Proof. 1.

$$\alpha \wedge \beta = \frac{(p+q)!}{p!q!} \operatorname{Alt}(\alpha \otimes \beta)$$

$$\beta \wedge \alpha = \frac{(p+q)!}{p!q!} \operatorname{Alt}(\beta \otimes \alpha)$$

Only need to show that

$$\operatorname{Alt}(\alpha \otimes \beta) = (-1)^{pq} (\beta \otimes \alpha)$$

Where

$$\operatorname{Alt}(\alpha \otimes \beta) = \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\operatorname{sgn} \sigma) \, \sigma(\alpha \otimes \beta)$$

Let $\tau = (p+1, \cdots, p+q, 1, 2, \cdots, p)$, then $\operatorname{sgn}(\tau) = (-1)^{pq}$. We have

$$
\begin{aligned}
\operatorname{Alt}(\alpha \otimes \beta) &= \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\operatorname{sgn}\sigma)\, \sigma \circ \tau (\beta \otimes \alpha) \\
&= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma \in S_{p+q}} (\operatorname{sgn}\sigma)(\operatorname{sgn}\tau)(\sigma \circ \tau)(\beta \otimes \alpha) \\
&= (-1)^{pq} \frac{1}{(p+q)!} \sum_{\sigma' \in S_{p+q}} (\operatorname{sgn}\sigma')\, \sigma' (\beta \otimes \alpha) \\
&= (-1)^{pq} \operatorname{Alt}(\beta \otimes \alpha)
\end{aligned}
$$

2. Suppose

$$
\begin{aligned}
\alpha &= \sum_{i_1 \cdots i_p} a_{i_1 \cdots i_p} e^*_{i_1} \wedge \cdots \wedge e^*_{i_p} \\
\beta &= \sum_{j_1 \cdots j_q} b_{j_1 \cdots j_q} e^*_{j_1} \wedge \cdots \wedge e^*_{j_q}
\end{aligned}
$$

Then

$$
\alpha \wedge \beta = \sum_{i_1 \cdots i_p} a_{i_1 \cdots i_p} b_{j_1 \cdots j_q} (e^*)_I \wedge (e^*)_J = \sum_{I,J} a_I b_J e_{IJ}
$$

$$
\beta \wedge \alpha = \sum_{I,J} a_I b_J e_{JI} = (-1)^{pq} \sum_{I,J} a_I b_J e_{IJ} = (-1)^{pq} \alpha \wedge \beta
$$

□

Exercise 0.1.2

Let V be a vector space of finite dimension and let V^* denotes the dual space. Prove that if $(\theta_1, \ldots, \theta_r)$ is a family of r covectors in V^* which are ***linearly dependent***, then the following wedge product:

$$
\theta_1 \wedge \cdots \wedge \theta_r = 0.
$$

Proof. Since $\theta_1 \wedge \cdots \theta_r$ is multi-linear and alternative, take any vector $v^1, \cdots, v^r$, we have

$$
\theta_1 \wedge \cdots \wedge \theta_r \left(v^1, \cdots, v^r\right) = \det\left(\theta_i\left(v^j\right)\right)_{ij}
$$

Since $\theta_1, \cdots, \theta_r$ are linealy dependent , the row vectors of the matrix $\left(\theta_i\left(v^j\right)\right)_{ij}$ are as well. Thus $\det\left(\theta_i\left(v^j\right)\right)_{ij} = 0$, and $\theta_1 \wedge \cdots \theta_r\left(v^1, \cdots, v^r\right) = 0$. Since $v^1, \cdots, v^r$ are arbitrary , $\theta_1 \wedge \cdots \wedge \theta_r = 0$. □

Exercise 0.1.3 ▶ Basis of Grassmann algebra

Let V be a vector space of finite dimension with basis $(e_1, \dots, e_n)$ and let V^* denotes the dual space with dual basis $(e_1^*, \dots, e_n^*)$. Please prove that

$$(e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*)_{1 \le i_1 < \cdots < i_k \le n}$$

forms a basis of $\Lambda^k V^*$.

Proof. □

Exercise 0.1.4 ▶ Grassmann algebra versus Polynomial functions on V.

Let V be a vector space of finite dimension and let V^* denotes the dual space. A p-multilinear map α on V is ***symmetric*** if

$$\alpha(v_1, \dots, v_n) = \alpha(v_{\sigma(1)}, \dots, v_{\sigma(n)})$$

for all permutations σ. Such α defines a homogeneous polynomial $\tilde{\alpha}$ defined by

$$\tilde{\alpha}(v) = \alpha(v, \dots, v)$$

for all $v \in V$. Then show that for symmetric multilinear maps α, β on V of respective p, q, the product $\alpha \circ \beta$ defined by

$$\alpha \circ \beta(v_1, \dots, v_{p+q}) := \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)})$$

is valued into symmetric multilinear map on V of degree $p + q$ and that

$$\alpha \circ \beta(v, \dots, v) = \alpha(v, \dots, v)\beta(v, \dots, v).$$

Could you compare with $\Lambda^\bullet V^*, \wedge$?

Exercise 0.1.5 ▸ Basic important examples of smooth manifolds

Please prove that the following topological spaces have a smooth manifold structure :

- The Euclidean sphere $S^n \subset \mathbb{R}^{n+1}$.
- The matrix groups $GL_n(\mathbb{R}), SL_n(\mathbb{R}), O_n(\mathbb{R})$, these are Lie groups of matrice. Each time could you tell me what is the dimension of these groups.

Exercise 0.1.6 ▸ Quotient manifolds

Prove that the following objects are quotient manifolds

- $\mathbb{R}P^n$ obtained by quotienting S^n by antipodal map, show that it is nonorientable by a picture. Could you explain by a picture that the neighborhood of the line at infinity is homeomorphic to a Möbius ribbon ?
- Prove that the torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$ is a smooth manifold.

Exercise 0.1.7 ▸ Vector fields are derivation

Let M be a C^∞ manifold. Show that the left C^∞ module of derivation of the ring $C^\infty(M)$ i.e. linear maps $D : C^\infty(M) \to C^\infty(M)$ satisfying

$$D(fg) = (Df)g + f(Dg)$$

can be identified with vector fields on M, smooth sections of TM.